| No. | User Stories | Acceptance Criteria | Story Points | | |
|---|---|---|---|---|---|
| US_03 | As a player, I want to be able to challenge an easy AI opponent so that I can practice basic gameplay mechanics. | An EasyController script is implemented and assigned to the AI. The AI selects actions such as block, jump, move, or attack at random. The AI moves in the general direction of the player. The AI has a slower reaction update time, making its behavior less responsive and easier to predict. | 10 | | |
| **No.** | **Test Case Title** | **Test Flow** | **Expected Result** | **Points Breakdown** | **Execution Status** |
| | | | | | |
| TC_03_01 | Easy AI Controller Loads | Launch the game and start a match selecting easy difficulty. Observe the AI behavior at the start of the match. | The EasyController is active and controls the AI opponent. | 3 | COMPLETE |
| TC_03_02 | Easy AI Performs Random Actions | Start a match on easy difficulty and observe the AI over time. | The AI performs actions such as block, jump, move, and attack in a random and simple manner. | 3 | COMPLETE |
| TC_03_03 | Easy AI Moves Toward Player | Start a match on easy difficulty and position the player away from the AI. Observe AI movement. | The AI moves generally in the direction of the player. | 2 | COMPLETE |
| TC_03_04 | Easy AI Has Slow Reaction Time | Start a match on easy difficulty and perform actions near the AI. Observe how quickly the AI responds. | The AI responds slowly, making it easier for the player to react and counter. | 2 | COMPLETE |
| US_04 | As a player, I want to be able to challenge a medium AI opponent so that I can experience a balanced level of difficulty. | A MediumController script is implemented and assigned to the AI. The AI selects actions such as block, jump, move, or attack with more consistency than random behavior. The AI moves toward the player more effectively than the easy AI. The AI has a moderate reaction update time, making it more responsive than easy but not overly difficult. | 10 | | |
| **No.** | **Test Case Title** | **Test Flow** | **Expected Result** | **Points Breakdown** | **Execution Status** |
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| TC_04_01 | Medium AI Controller Loads | Launch the game and start a match selecting medium difficulty. Observe the AI behavior at the start of the match. | The MediumController is active and controls the AI opponent. | 3 | COMPLETE |
| TC_04_02 | Medium AI Performs More Consistent Actions | Start a match on medium difficulty and observe the AI over time. | The AI performs actions more consistently and less randomly than easy difficulty. | 3 | COMPLETE |
| TC_04_03 | Medium AI Tracks Player Movement | Start a match on medium difficulty and move the player around the stage. Observe AI movement. | The AI follows and positions itself toward the player more effectively than easy AI. | 2 | COMPLETE |
| TC_04_04 | Medium AI Has Moderate Reaction Time | Start a match on medium difficulty and perform actions near the AI. Observe response timing. | The AI responds faster than easy difficulty but not instantly. | 2 | COMPLETE |
| US_05 | As a player, I want to be able to challenge a hard AI opponent so that I can face a highly challenging and intelligent opponent. | A HardController script is implemented using machine learning techniques. Unity ML-Agents are installed and configured with a working Python environment. The AI overrides CollectObservations, OnActionReceived, and OnEpisodeBegin methods. A reward and penalty system is implemented to guide AI learning. The AI is trained using a Python training process. The resulting trained neural network model is imported and used in the game. The hard AI demonstrates advanced and responsive behavior compared to easy and medium difficulties. | 15 | | |
| **No.** | **Test Case Title** | **Test Flow** | **Expected Result** | **Points Breakdown** | **Execution Status** |
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| TC_05_01 | Hard AI Controller and Model Load | Launch the game and start a match selecting hard difficulty. Observe the AI at the start of the match. | The HardController and trained neural network model are loaded and active. | 3 | COMPLETE |
| TC_05_02 | Hard AI Demonstrates Advanced Behavior | Start a match on hard difficulty and observe the AI during gameplay. | The AI shows intelligent behavior such as well-timed attacks, blocking, and movement. | 5 | COMPLETE |
| TC_05_03 | Hard AI Reacts Quickly to Player Actions | Start a match on hard difficulty and perform various actions near the AI. | The AI responds quickly and adapts to player actions with minimal delay. | 2 | COMPLETE |

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| TC_05_04 | Hard AI Training Integration Works | Launch the game after importing the trained model and play a match on hard difficulty. | The AI behavior reflects the trained model and runs without errors or missing functionality. | 5 | COMPLETE |
| US_11 | As a player, I want the camera to keep both fighters visible during the match so that I can clearly see the action and react to my opponent. | The camera view displays both the player and AI opponent during the fight. When fighters move across the stage, the camera adjusts its position to keep both characters visible on screen. The camera does not cut off characters at the screen edges during normal gameplay. Camera movement is smooth and does not cause abrupt jumps or disorienting transitions. | 6 (2 already earned in sprint 1) | | |
| **No.** | **Test Case Title** | **Test Flow** | **Expected Result** | **Points Breakdown** | **Execution Status** |
| TC_11_01 | Camera Shows Both Fighters | Start a match and observe camera positioning during gameplay. | Both fighters remain visible on screen at the same time. | 3 | SPRINT 3 |
| TC_11_02 | Camera Follows Fighter Movement | Move fighters across the stage to opposite ends. | Camera adjusts position to keep both fighters visible. | 3 | SPRINT 3 |
| US_14 | As a player, I want to be able to select between multiple difficulties so that I can choose a gameplay experience that matches my skill level. | The game provides an option to select between easy, medium, and hard difficulties before starting a match. The selected difficulty correctly loads the corresponding AI controller. The selection is clearly visible to the player. The game starts with the chosen difficulty applied. | 4 | | |
| **No.** | **Test Case Title** | **Test Flow** | **Expected Result** | **Points Breakdown** | **Execution Status** |
| TC_14_01 | Difficulty Selection Menu Displays | Launch the game and navigate to the mode or match setup screen. | The player is presented with options to select easy, medium, or hard difficulty. | 2 | SPRINT 3 |
| TC_14_02 | Selected Difficulty Persists Into Match | Launch the game, select any difficulty, and start a match. Observe gameplay settings. | The match runs using the selected difficulty without reverting or changing unexpectedly. | 1 | SPRINT 3 |
| TC_14_03 | Difficulty Selection Loads Correct AI | Launch the game, select difficulty, and start a match. Observe AI behavior. | The AI behaves according to easy, medium, and hard difficulty. | 2 | SPRINT 3 |